

MediTap

Technical Q&A Brief

REPORT DATE	August 27, 2026	DOCUMENT	Technical Q&A Brief
AUDIENCE	Founders / advisors / leadership	STATUS	As-built product snapshot
SCOPE NOTE	Local Docker + feature/portal-split	COMPANION MD	meditap-technical-qa-brief-2026-08-27.md

BOTTOM LINE

MediTap today is a working dual-portal chart product: patient user portal (appointments, labs, insurance, conditions, incidents, meds/allergies, intake) plus clinic admin portal (patients, charts, hospitals, documents, activity), with multi-EHR PDF intake as a chart-feeding capability. Not yet a production EHR connector, HIPAA certification claim, or LLM clinical engine.

0 Executive snapshot

Topic	One-line answer
What exists	User + admin portals; live chart APIs; PDF intake
User portal	Dashboard, appts, labs, insurance, conditions, intake
Admin portal	Staff login, patients, charts, hospitals, docs
Not done yet	Prod EHR write-back; full audit; live NFC registry
OCR / LLMs	Tesseract.js yes; no LLM in intake
Safe claims	No HIPAA cert / clinical % / live EHR sync yet

1 Current product / what we've built

Q: What have we actually built and successfully tested?

Lead with the working product - not only recent PDF fixtures.

User portal (patient-facing)

- Dashboard + Quick Status with next steps
- Appointments - cards/modal + Django appointments API
- Chronic conditions + incidents/hospital visits (API + staff quick-pick)
- Lab results (PatientLabPanel API) + insurance/payers
- Patient intake with medications & allergies; Save to API
- Settings; JWT login; staff elevation for kiosk edits

Admin portal (clinic staff - working, not a stub)

- Staff login door; patients blocked from admin routes
- Patient search/hub + clinical charts + embedded patient view
- On-behalf intake, labs, appointments, insurance for selected patient
- Hospitals CRUD; document vault/review queue; activity trail
- Epic sandbox panel (partial - not production sync)

PDF intake (feeds the chart; not the whole product)

EHR source selector + pdf.js/Tesseract + Athena/MEDITECH/Epic/Generic parsers + Accept/Reject gates. Fixture corpora (>=90-95% section completeness on locked demo PDFs) are a parser regression loop - they do not define the whole product.

Q: Functional vs planned?

Functional today	Partial / in progress	Planned / do not claim
User portal chart surfaces	Portal polish	Prod multi-tenant EHR sync
Admin portal ops shell	On-behalf polish / queues	Full HIPAA audit product
Chart CRUD + staff elevation	Tab14 API-only cleanup	LLM understanding
PDF/OCR + dialects	NextGen + broader corpus	Live NFC card registry
Document vault v1	Apply-to-chart polish	Certified EHR write-back

Q: Languages?

TypeScript/JavaScript (SPA + parsers), Python (Django), SQL (Postgres), Docker/shell.

Q: Frameworks / libraries?

React 19, Ionic React 8, Vite, React Router, i18next, pdf.js, Tesseract.js; Django + DRF + SimpleJWT; Vitest/ESLint; Capacitor packaging path.

Q: Tech stack?

Layer	Choice
Frontend	Ionic React + Vite SPA
Backend	Django REST (medapp + medical)
Database	PostgreSQL 16
Auth	Django JWT + staff elevation token
Files	PatientDocument FileField + parse_snapshot JSON
Hosting (dev/demo)	Docker Compose; GCP VM notes for demos
OCR / AI	Tesseract.js / no LLM in intake

Q: What does each technical team member handle?

In-repo ownership highlights Antonio for product continuity / intake direction. Replace with a named org chart before external investor use: frontend (meditap-app), backend (Django), infra/Docker as assigned.

Q: What does the backend actually do? (simple)

ELEVATOR ANSWER

The backend is MediTap's system of record and security gate: it authenticates patients and staff, stores structured chart data (appointments, labs, meds/allergies, insurance, conditions, incidents, demographics, documents), and exposes REST APIs both portals use. PDF understanding mostly runs in the browser; the API persists results.

2 Medical record upload & processing

PDF intake feeds the working chart. Appointments, labs, insurance, and other portal features already work as first-class API-backed surfaces with or without an upload.

Q: What happens on upload?

- Select EHR source on Tab14.
- Upload PDF or image.

- Browser extracts text (pdf.js); OCR sparse/image pages (Tesseract.js).
- Dialect/general parsers fill structured fields.
- User Accepts/Rejects flagged imports; optional completeness hints.
- Save / document APIs persist to Postgres (+ keep original file in vault).

Q: How do we extract? OCR? LLMs?

Native PDF text when available; Tesseract.js OCR for images/sparse pages; vendor-specific heuristic parsers. No ChatGPT-style LLMs in the intake path today.

Q: What can we extract?

Demographics, care team, allergies, meds, problems, labs/results, vitals, procedures, immunizations, encounters, payers, social/functional/mental status, histories, notes, PoT, and Athena-oriented extras when present - quality depends on the export.

Q: Who decides meaning / storage / originals?

Frontend parsers map text to MediTap fields; backend validates auth and stores typed models in PostgreSQL. Originals retained as PatientDocument (not auto-deleted). Production retention policy is still a compliance design item.

3 Accuracy / validation

Q: How do we keep extraction accurate?

Vendor parsers + fixture corpus tests + Accept/Reject gates + staff-only clinical edits. Blurry/handwritten/odd layouts degrade OCR; unsure fields stay empty or flagged.

Q: Confidence scores?

Today: section completeness / portability coverage scores - not clinical per-field probability. True confidence scoring is optional future work.

Q: Human verification before EHR?

Yes in product design (patient + staff review). EHR write-back is not live, so nothing auto-enters Epic/Cerner/athena today.

Q: What accuracy numbers can I safely mention?

SAFE WORDING

On locked demo fixtures, automated section-completeness gates are typically $\geq 90\text{-}95\%$ for Athena / MEDITECH / Epic training PDFs. Do NOT claim published clinical accuracy, FDA clearance, or physician-grade charting quality.

4 EHR integration / future

Q: What still needs to be built for EHR integration?

Production OAuth/SMART, FHIR client + tokens on the MediTap server, resource mapping, sync/conflict/provenance, BAAs and customer enablement, reviewed write-back UI. See docs/epic-fhir-meditap-integration-report.md.

Q: FHIR in simple terms?

A shared web language for health data - standardized JSON-style resources for patient, meds, labs, allergies - so apps and EHRs exchange chart pieces without a private format each time.

Q: Other standards? Convert to FHIR before send?

Also SMART/OAuth, CCD/C-CDA (already parsed), and planned terminology (RxNorm/SNOMED/LOINC). For FHIR EHRs: yes - map MediTap models to FHIR resources before API send. Intake today maps PDF to MediTap models first.

Q: One integration or many?

Shared architecture with per-vendor adapters. Needs practice/EHR permission. Barriers: enablement, scopes, identity binding, write limits, capability variance, PHI liability.

Q: Why can't Epic already do this? / vs uploading a PDF?

EHRs excel inside their network; patients still arrive with outside PDFs/CCDs. Uploading a PDF is often a file blob. MediTap aims for discrete, reviewable fields and a later FHIR handoff - not an unparsed attachment.

5 Security / patient data

Q: How is PHI protected today?

API authz, staff elevation, portal split. Production needs HTTPS, rotated secrets, hosting controls. Transit: TLS in real deploys. At rest: rely on host/cloud disk encryption when configured - not 'every column HSM-encrypted' as a current claim.

Q: What/where stored? Auth? Audit? Card?

Structured chart + documents in MediTap Postgres/media. Patients use JWT accounts; staff use admin login and/or elevation. Starter AdminActivityEvent exists - not a full HIPAA audit product. NFC/RFID card should identify/authenticate only; PHI stays on servers. Card lost/inactive UI is demo status today - not a live kill registry.

DO NOT CLAIM YET

HIPAA certified/compliant without counsel attestation; fully integrated with Epic/Cerner/athena; AI diagnoses; clinical extraction accuracy % beyond fixtures; PHI on the physical card; live multi-clinic card kill-switch.

6 Competitive / technical moat

Q: What is hard / unique / hardest to copy?

Multi-vendor dialect engineering under real export chaos; patient-mediated intake with Accept/Reject + staff elevation; path from documents to structured chart to future FHIR. Hardest to copy: accumulated parsers + regression corpora + workflow gates - not 'call OCR on a PDF.'

Q: Biggest technical advantage?

ENGINEERING VIEW

A working dual portal (patient + admin) with live chart APIs, plus the ability to bootstrap that chart from messy real EHR exports across major vendors - with humans in the loop and a clear FHIR path. Harder than a PDF vault or a FHIR hello-world alone.

7 30-second close

- Working user portal + working admin portal today - appts, labs, insurance, conditions, meds/allergies, intake.
- PDF intake feeds that chart: upload -> pdf.js + Tesseract -> vendor parsers -> Accept/Reject -> Postgres.
- No LLMs in extraction today; FHIR is the planned EHR bridge.
- Accuracy guarded by fixtures + humans - not marketing percentages.
- Demo completeness != clinical certification; sandbox != production EHR.